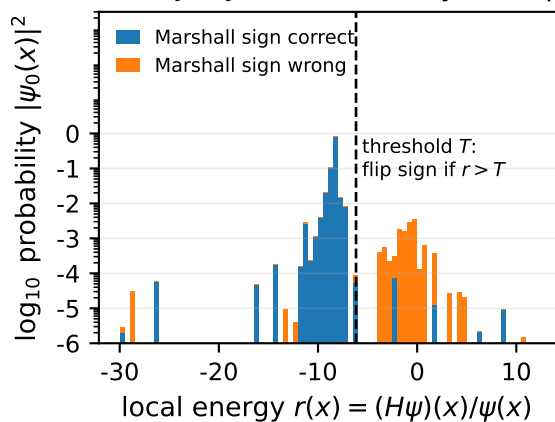
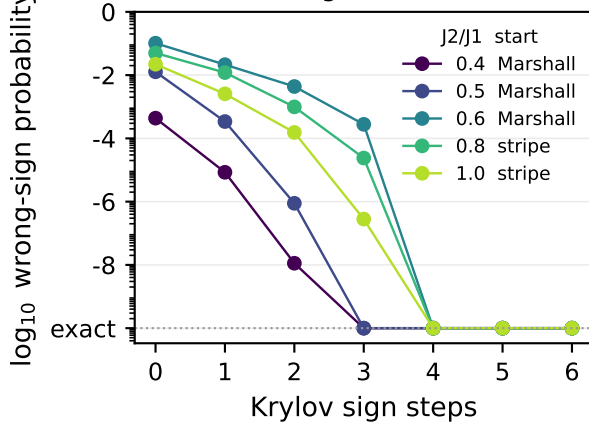
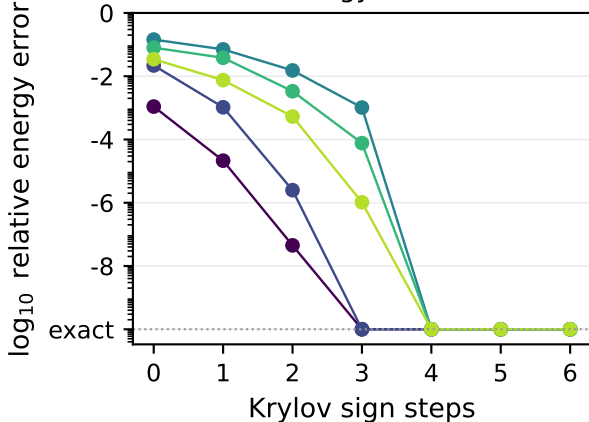


(a) 4×4 , $J_2 / J_1 = 0.5$: first Krylov step**(b)** 4×4 : sign error vs ED**(c)** 4×4 : energy error vs ED**(d)** $J_2 / J_1 = 0.5$: two exact clusters